

this problem the author used the limit of small β where $Y_{\nu,m}(\theta, \phi) \sim J_\nu[(\nu + \frac{1}{2})\theta]e^{im\phi}$ and showed that the coupling of a wave into a conical wave guide implies effective dipoles given by the Bethe-Bouwkamp solution. These dipoles are perpendicular to the aperture, a result which is completely different from the problem of radiation by a tip which generates dipoles located in the aperture plane.

For completeness, we wish to comment on some additional, interesting properties of the emission by conical tips that are relevant to the experiment. The quasi-multipole model shows that the aperture radius disappears in the formula Eq. 17 and, consequently, in the radiation profile. We can understand this fact using the analogy with Rayleigh's and Mie's theory[12] of diffusion by small particles which follow the same behavior as our model and which can be associated with the impossibility to see details smaller than the wavelength of light. In particular, the aperture size cannot be a relevant variable in the limit of small apertures. For the same reason, the tip angle β does not play an important role in the diffraction profile. The dependence on the aperture radius is very different in the short-wavelength limit and for large apertures $a \geq \lambda$. For a large hole, the transmission must approach unity and, by consequence, the aperture radius may appear in the profile. We have then $\frac{dP}{Pd\Omega} \sim (ka)^2 \cos^2(\theta) \left(\frac{2J_1(ka \sin(\theta))}{ka \sin(\theta)} \right)^2$ and $P \sim cE_0^2 a^2 / 8$ that can be compared with $P \sim \frac{ck^6 a^6}{k^2} \mathbf{E}_0^2$, valid for Bethe's theory and our conical subwavelength aperture model.

VII. SUMMARY

An optical fiber tip with a small aperture in the metal coating as used in NSOM is usually characterized by its angular transmission profile. In order to describe this profile theoretically, we have presented in this paper an analytical model for light diffraction through a subwavelength aperture located at the apex of a metallic screen with conical geometry. For this purpose, a field expansion in quasi spherical multipoles has been developed to solve Maxwell's equations analytically. Special care has been taken to consider realistic boundary conditions adapted not only to the conical geometry but also to the finite conductivity of the coating. Our model is able to reproduce and to explain the experimental results. The diffraction in the far-field is demonstrated to remain similar to that drawn with the 2-dipole model[5, 6] over a large portion of space. The latter model is formally justified in our approach. In particular, we find a characteristic factor of 2 between the expressions for